

Properties of Eigenvectors

Understanding special directions, transformations, and AI/ML applications — a classroom guide for 3rd-semester BTech students in Data Science & Machine Learning.

LINEAR ALGEBRA · DATA SCIENCE · ML FOUNDATIONS



Quick Recap: What Is an Eigenvector?

The Core Equation

$$Av = \lambda v$$

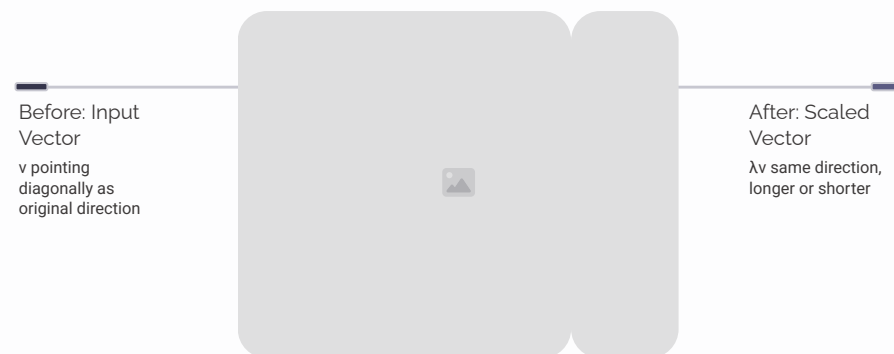
A – the transformation matrix that acts on a vector

v – the eigenvector: a *special direction* in space

λ (lambda) – the eigenvalue: a scalar that tells us how much the vector is scaled

□ Intuition: After multiplying by A, the eigenvector v points in the exact same (or opposite) direction – only its length may change.

Geometric Picture



Most vectors *rotate* after a transformation. An eigenvector is special – it only **stretches or shrinks**, never rotates.

Property 1: The Eigenvector Cannot Be Zero

Why $v \neq 0$?

$$v \neq \mathbf{0}$$

The zero vector satisfies $\mathbf{A}v = \mathbf{0} = \lambda \cdot \mathbf{0}$ for *every* matrix A and every scalar λ . It tells us nothing special about A — it fits everywhere and therefore identifies no unique direction.

 The whole point of an eigenvector is to find a **specific direction** that the matrix preserves. The zero vector has no direction at all.

Simple 2D Example

Let $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$

If we try $v = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ (zero vector):

$$Av = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This is trivially true for *any* matrix — zero is never accepted as an eigenvector.



KEY LEARNING: EIGENVECTORS MUST ALWAYS BE NON-ZERO VECTORS.

Property 2: Scalar Multiples Are Also Eigenvectors

The Math

If $Av = \lambda v$, then for any non-zero scalar c :

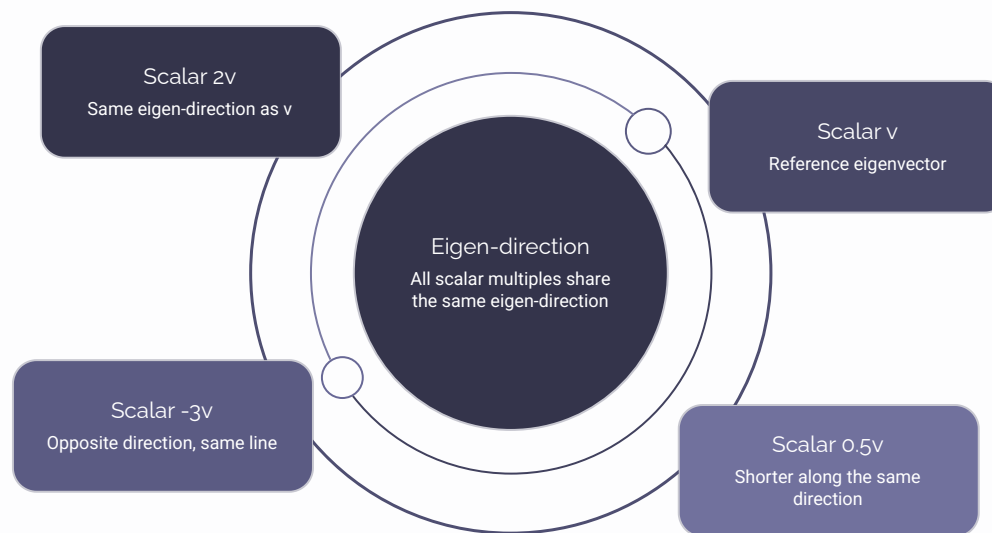
$$A(cv) = c(Av) = c(\lambda v) = \lambda(cv)$$

So cv is also an eigenvector with the same eigenvalue λ .

Example

Let $v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Then $2v, 5v, -3v$ all point along the x-axis – they are all valid eigenvectors for the same eigenvalue!

Visual: The Eigen-Direction Line



Key Learning: Eigenvectors are really about **directions**, not specific vector lengths. Infinitely many vectors share one eigen-direction.

Property 3: Different Eigenvalues $\rightarrow$ Linearly Independent Eigenvectors

The Rule

Eigenvectors that correspond to **distinct eigenvalues** are always **linearly independent** – meaning none of them can be written as a combination of the others.

Worked Example

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

$$\lambda_1 = 2 \rightarrow v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ (x-direction)}$$

$$\lambda_2 = 3 \rightarrow v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \text{ (y-direction)}$$

You cannot express v_1 as a multiple of v_2 or vice versa – they are independent directions.

Intuition

Different Eigenvalue

Each eigenvalue "scales" its eigenvector by a unique factor – so each eigenvector points in a genuinely different direction.

Why It Matters

Having many independent eigenvectors means we can decompose any vector in the space – powerful for PCA and spectral methods in ML.



Key Learning: Distinct eigenvalues always guarantee independent eigenvectors – no overlap in directions.

Property 4 & 5: Length Doesn't Matter — But Normalization Helps

Property 4 — Any Length Works

An eigenvector can have **any non-zero magnitude**. Consider:

$$v = \begin{bmatrix} 10 \\ 0 \end{bmatrix}$$

If $A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$, then:

$$Av = \begin{bmatrix} 40 \\ 0 \end{bmatrix} = 4 \cdot \begin{bmatrix} 10 \\ 0 \end{bmatrix} = \lambda v \checkmark$$



BEING AN EIGENVECTOR $\neq$ BEING A UNIT VECTOR

Property 5 — Normalizing an Eigenvector

$$q = \frac{v}{\|v\|}$$

Example: Let $v = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$

$$\|v\| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$$

$$q = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix} = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$$

Now $\|q\| = 1$. This normalized vector is still an eigenvector but has unit length — crucial for **PCA** and numerical algorithms.

Property 6: What Does the Eigenvalue λ Tell Us?

The eigenvalue controls how the eigenvector is scaled after transformation. Here's a complete visual guide:

Condition	Effect on v	Geometric Meaning	ML Example
$\lambda > 1$	Vector gets longer	Stretching in that direction	High-variance PCA component
$0 < \lambda < 1$	Vector gets shorter	Shrinking / compression	Low-variance component
$\lambda = 1$	No change at all	Identity in that direction	Preserved feature axis
$\lambda = 0$	Mapped to zero	Direction collapsed – matrix singular	Dropped PCA component
$\lambda < 0$	Direction flips + scales	Reflection + stretch/shrink	Sign-reversal in transformation

  **Key Learning:** The eigenvalue is not just a number — it tells a complete story about what the matrix does to that special direction.

Property 7 & 8: Orthogonal and Orthonormal Eigenvectors

Property 7 — Orthogonality for Symmetric Matrices

If $A = A^T$ (symmetric), eigenvectors for **different eigenvalues** are always perpendicular:

$$v_1^T v_2 = 0$$

Example: $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$v_1^T v_2 = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0 \checkmark$$

They point along the x and y axes — perfectly perpendicular!

Property 8 — Orthonormal = Orthogonal + Unit Length

Orthogonal

$v_i^T v_j = 0$ for $i \neq j$. Vectors point in perpendicular directions.

Orthonormal

Orthogonal **AND** each vector has length 1: $\|v_i\| = 1$.

Why it matters in ML: PCA computes orthonormal eigenvectors of the covariance matrix. Orthonormality ensures the principal components are independent axes with no redundancy — making dimension reduction clean and interpretable.

🔑 SYMMETRIC MATRICES → ORTHOGONAL EIGENVECTORS (GUARANTEED!)

Properties 9 & 10: Eigenvectors of A^2 , A^{-1} , and A^k

Property 9 — A^2 Has Same Eigenvectors

Start from $Av = \lambda v$, apply A again:

$$A^2v = A(Av) = A(\lambda v) = \lambda(Av) = \lambda^2v$$

Example: If $Av = 3v$, then:

$$A^2v = 9v$$

The direction is unchanged — only the scale is squared!

Property 10 — A^{-1} Flips the Eigenvalue

From $Av = \lambda v$, multiply both sides by A^{-1} :

$$v = \lambda A^{-1}v$$

$$A^{-1}v = \frac{1}{\lambda}v$$

Example: If $\lambda = 4$, then A^{-1} has eigenvalue $\frac{1}{4}$.
Same eigenvector, reciprocal eigenvalue!



Eigenvector stays the same;
eigenvalue becomes its reciprocal.

Property 11 — General Power A^k

$$A^k v = \lambda^k v$$

Example: If $Av = 2v$:

- $A^2v = 4v$
- $A^3v = 8v$
- $A^{10}v = 1024v$

This powers up the scale — but the eigenvector always stays pointing in the same direction.
Used in **graph algorithms and PageRank** in ML!

Eigenvectors and PCA — The Most Important ML Connection

How PCA Uses Eigenvectors

PCA (Principal Component Analysis) is built entirely on eigenvectors of the **covariance matrix** of your data:

01

Compute the Covariance Matrix

Captures how features vary together across data points.

02

Find Eigenvectors and Eigenvalues

Eigenvectors = principal directions of variation in the data.

03

Rank by Eigenvalue Size

Larger eigenvalue = more variance = more important direction.

04

Keep Top k Components

Discard low-variance directions → reduce dimensions without losing much information.

Worked Intuition

Suppose your covariance matrix has eigenvalues:

$$\lambda_1 = 100, \quad \lambda_2 = 20, \quad \lambda_3 = 2$$

Total variance = 122. The first two components capture:

$$\frac{100 + 20}{122} \approx 98.4\%$$

PCA drops the third direction ($\lambda_3 = 2$) — only 1.6% of variance lost. We go from 3D to 2D with almost no information loss!

✅ 🔑 Eigenvectors give the directions; eigenvalues tell you how important each direction is.

AI/ML Applications of Eigenvectors

Eigenvectors appear across virtually every major area of machine learning and data science:



PCA & Dimensionality Reduction

Eigenvectors of the covariance matrix define principal components. Reduce 100s of features to just 2–3 while preserving most variance.



Image Compression (SVD)

Singular value decomposition uses eigenvector-like concepts to compress images by keeping only the most important directions of variation.



Recommendation Systems

Matrix factorization decomposes user-item matrices using eigenvectors to discover latent features (e.g., genre preferences).



Spectral Clustering

Graph Laplacian eigenvectors are used to partition data into clusters — powerful for community detection in social networks.



Google PageRank

PageRank is the dominant eigenvector of the web link matrix — the most "important" direction in the web graph structure.



Face Recognition (Eigenfaces)

Eigenvectors of a face image dataset form "eigenfaces" — compact basis directions that represent the most common facial variations.

Complete Worked Example

The Matrix

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$$

Eigenvector 1: $v_1 = [1, 0]^T$

$$Av_1 = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

✓ $\lambda_1 = 4$ — the x-direction is stretched by a factor of 4.

Eigenvector 2: $v_2 = [0, 1]^T$

$$Av_2 = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

✓ $\lambda_2 = 2$ — the y-direction is stretched by a factor of 2.

Visual Summary



x-direction

Stretches $\times 4$. Direction preserved.

y-direction

Stretches $\times 2$. Direction preserved.

- ☐ Both eigenvectors are orthogonal and independent — confirming Properties 3 and 7.

Common Student Mistakes to Avoid

✗ Mistake 1: Zero Vector as Eigenvector

The zero vector satisfies $Av = \lambda v$ for every matrix — it is explicitly **excluded by definition**. Always ensure $v \neq 0$.

✗ Mistake 2: Thinking Eigenvectors Must Have Length 1

Any non-zero scalar multiple of an eigenvector is also an eigenvector. Length 1 (normalized) is a **choice for convenience**, not a requirement.

✗ Mistake 3: Confusing Eigenvalue with Eigenvector

The **eigenvalue λ** is a scalar (a number). The **eigenvector v** is a vector (a direction). They answer different questions — "how much?" vs. "which direction?"

✗ Mistake 4: Forgetting Scalar Multiples Are Also Eigenvectors

If v is an eigenvector, so is $2v$, $-5v$, and $100v$. An eigenvector represents a whole **line of directions**, not just one specific vector.

✗ Mistake 5: Assuming Every Matrix Has Orthogonal Eigenvectors

Orthogonality is **only guaranteed for symmetric matrices** ($A = A^T$) with distinct eigenvalues. General matrices need not have orthogonal eigenvectors.

Quick Revision — All Properties at a Glance

#	Property	Meaning	Key Example / Formula
1	Non-zero vector	$v \neq 0$ always; zero vector excluded by definition	$v = [0,0]^T \rightarrow$ rejected
2	Scalar multiples	cv is also an eigenvector for same λ ($c \neq 0$)	$v, 2v, -3v$ all valid
3	Linear independence	Different eigenvalues $\rightarrow$ linearly independent eigenvectors	$\lambda_1 \neq \lambda_2 \rightarrow v_1, v_2$ independent
4	Any magnitude	Eigenvectors can have any non-zero length	$[10, 0]^T$ is as valid as $[1, 0]^T$
5	Normalization	$q = v/\ v\ $ gives unit eigenvector	$[3,4]^T \rightarrow [3/5, 4/5]^T$
6	Eigenvalue meaning	λ controls stretch, shrink, flip, or collapse	$\lambda > 1$ stretch; $\lambda < 0$ flip
7	Orthogonality	Symmetric $A \rightarrow$ eigenvectors $\perp$ for distinct λ	$v_1^T v_2 = 0$
8	Orthonormal	Orthogonal + unit length — used in PCA	$\ v_i\ =1$ and $v_i^T v_j=0$
9	Power A^2	$A^2v = \lambda^2v$ — eigenvalue squares, direction same	$Av=3v \rightarrow A^2v=9v$
10	Inverse A^{-1}	$A^{-1}v = (1/\lambda)v$ — eigenvalue becomes reciprocal	$\lambda=4 \rightarrow \lambda^{-1}=1/4$
11	General power A^k	$A^kv = \lambda^kv$ — same direction, eigenvalue raised to k	$Av=2v \rightarrow A^3v=8v$

Practice Problems

 EASY

Problem 1

Given $A = \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}$ and $v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

Verify that v is an eigenvector of A and find its eigenvalue.

Problem 2

If $v = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$ is an eigenvector with $\lambda = 3$, is $w = \begin{bmatrix} 10 \\ 0 \end{bmatrix}$ also an eigenvector? What is its eigenvalue?

 MEDIUM

Problem 3

Normalize the eigenvector $v = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$. Show your calculation step by step.

Problem 4

A symmetric matrix A has eigenvalues $\lambda_1 = 7$ and $\lambda_2 = 2$ with eigenvectors v_1 and v_2 . Can v_1 and v_2 be parallel? Justify your answer using two properties.

 ML / APPLIED

Problem 5

A covariance matrix has eigenvalues 200, 50, and 3. What percentage of variance is retained if you keep only the top 2 principal components? Should you keep the third?

Problem 6

If the dominant eigenvector of a transformation matrix A has eigenvalue $\lambda = 0.1$, what happens to that direction after applying A 10 times (i.e., $A^{10}v$)? Is this direction amplified or suppressed?

 Try each problem independently before moving to the next slide for answers!

Answers and Key Learnings

1

$$Av = [3, 0]^T = 3 \cdot [1, 0]^T \rightarrow \lambda = 3 \checkmark$$

Learning: Diagonal matrices scale each axis separately — each standard basis vector is automatically an eigenvector.

2

$$\text{Yes — } w = 5v, \text{ and } A(5v) = 5(Av) = 5 \cdot 3v = 3 \cdot (5v), \text{ so } \lambda = 3 \checkmark$$

Learning: Scalar multiples always share the same eigenvalue. Direction, not length, defines the eigen-direction.

3

$$\|v\| = \sqrt{36+64} = 10 \rightarrow q = [0.6, 0.8]^T \checkmark$$

Learning: Normalization = divide by magnitude. The result is a unit eigenvector — same direction, length = 1.

4

No — distinct eigenvalues ($7 \neq 2$) $\rightarrow v_1 \perp v_2$ (Property 7) and linearly independent (Property 3)

Learning: Symmetric matrices with distinct eigenvalues guarantee orthogonal eigenvectors — they cannot be parallel.

5

Variance retained = $(200+50)/253 \approx 98.8\%$ — the third component ($3/253 \approx 1.2\%$) can safely be discarded.

Learning: In PCA, compare eigenvalues to decide how many components to keep. Small eigenvalue = negligible variance.

6

$$A^{10}v = (0.1)^{10} v = 10^{-10} v \text{ — essentially zero. This direction is completely suppressed.}$$

Learning: $\lambda < 1 \rightarrow$ repeated application shrinks the direction exponentially. Used in iterative solvers and stability analysis.

Final Takeaways



Eigenvector = Special Direction

A direction that survives a matrix transformation unchanged — only scaled, never rotated.



Eigenvalue = Scale of That Direction

λ tells you whether the direction stretches, shrinks, stays, flips, or collapses to zero.

∞ Scalar Multiples = Same Direction

An infinite family of vectors share every eigen-direction. We usually pick the unit-length one.

$\perp$ Symmetric $\rightarrow$ Orthogonal

Symmetric matrices guarantee perpendicular eigenvectors for distinct eigenvalues — the foundation of PCA.



Master eigenvectors — and you hold the key to understanding how data science and machine learning truly work.